A 10-mW 16-b 15-MS/s Two-Step SAR ADC With 95-dB DR Using Dual-Deadzone Ring Amplifier

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***Abstract*— Low-noise ring amplifiers required for high-precision analog–digital converters (ADCs) greater than 16 b remain unexplored. This article demonstrates a two-step successive approximation (SAR) ADC achieving 91-dB signal-to-noise-and-distortion-ratio (SNDR) with 6-V differential input resulting in a low-frequency Schreier-figure-of-merit ($FOM_{S,lf}$) of 179.8 dB at 15 MS/s. The state-of-the-art performance is enabled by the ring-amplifier design that enables low noise during amplification and robust control of the transient dynamics. The ADC also features an on-chip residue amplifier (RA) settling characterization using the backend SAR ADC. The ADC is fabricated in a 180-nm CMOS process and occupies an active area of 1.82 mm^2.**

***Index Terms*— 16-bit ADC, analog–digital converter (ADC), low power, ring amplifier (RAMP), two-step SAR ADC.**

I. Introduction

SUCCESSIVE-APPROXIMATION register (SAR) analog-digital converters (ADCs) have been shown to achieve excellent power efficiencies due to their simple building blocks that are digital in nature and faster conversion rates as they scale favorably with technology. High-resolution ADCs [1] with stringent noise requirements have led to the adoption of hybrid ADC architectures such as the two-step SAR [2], which combine the concepts of SAR and pipeline ADC architectures.

Fig. 1 shows the accuracy requirement during the conversion of: 1) binary SAR; 2) SAR with redundancy; and 3) two-step SAR with redundancy. As conversion proceeds from most significant bit (MSB) to least significant bit (LSB) decisions, the residue reduces from the input-signal level to the LSB level. For a binary SAR, the conversion accuracy in terms of digital-to-analog converter (DAC) settling, reference, and comparator noise is required to be at the full ADC accuracy, which translates to longer bit conversion times.

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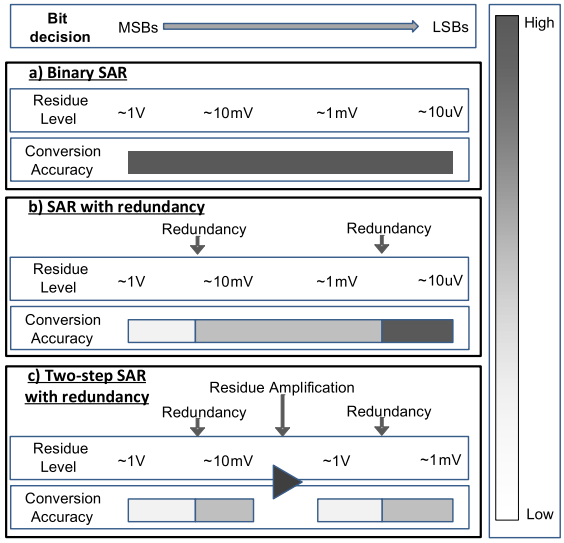


Fig. 1. Conversion accuracy requirements for (a) binary SAR, (b) SAR with redundancy, and (c) two-step SAR with redundancy.

Introducing redundancy in Fig. 1(b) allows early decision errors to be absorbed into the redundancy range of later decisions, shifting the full ADC accuracy requirement to the later LSB decisions. Finally, for a two-step SAR with redundancy [Fig. 1(c)], by introducing residue amplification, the backend SAR processing the LSBs is further relaxed by the gain introduced. By shifting the full ADC accuracy requirement to a single residue amplification step, the entire SAR operation consisting of the front-end and the back-end SAR can be optimized for fast conversion and low power.

An amplifier architecture well suited for residue amplification known as the ring amplifier (RAMP) [3] has enabled power-efficient two-step SAR ADCs. However, RAMP-based ADCs with signal-to-noise-and-distortion-ratio (SNDR) greater than 80 dB have not been attempted in previous literature [4]–[10]. In this work, the RAMP was designed to provide residue amplification for a high-resolution two-step SAR ADC achieving a DR of 95 dB and a peak SNDR of 91 dB. The ADC implemented in 0.18-μm technology consumes a total of 9.8 mW at 15 MS/s from 1.8-V and 3.3-V supplies, and occupies an area of 1.82 mm^2.

Section II discusses the system details of the ADC. Section III introduces the RAMP and describes the motivation and design details of the dual-deadzone (DDZ) RAMP. Finally,

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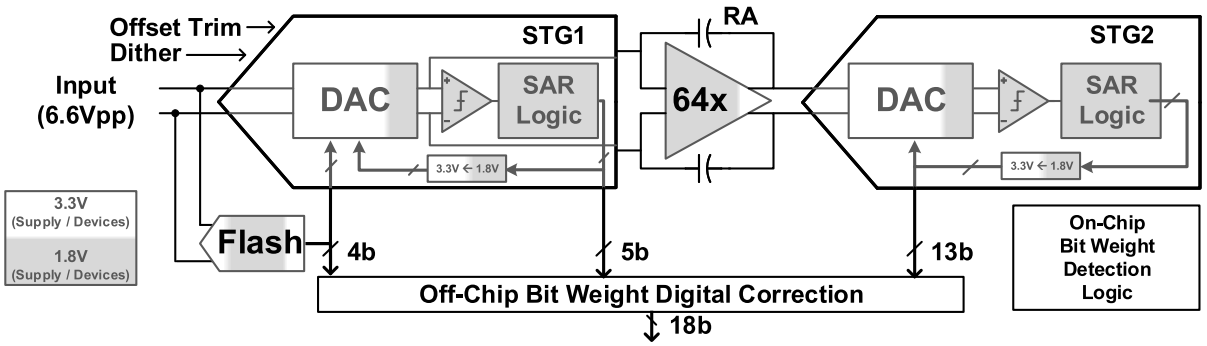
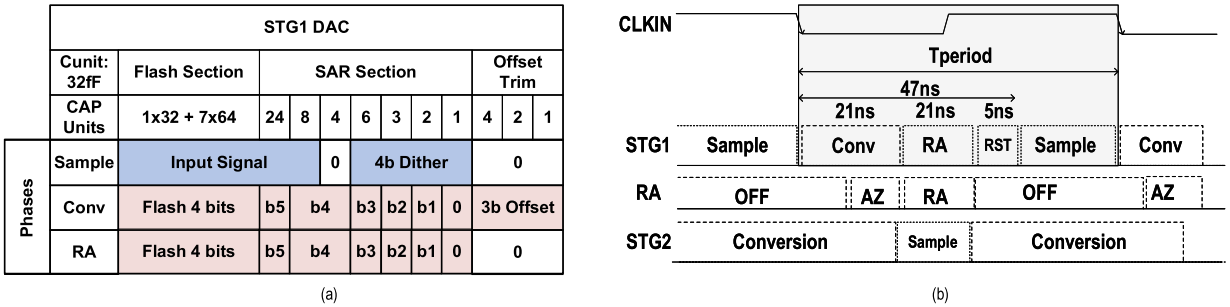


Fig. 2. Implemented two-step SAR ADC architecture.



		STG1 DAC										
	Cunit: 32ff	Flash Section	SAR Section							Offset Trim		
	CAP Units	1x32 + 7x64	24	8	4	6	3	2	1	4	2	1
Phases	Sample	Input Signal			0	4b Dither				0		
	Conv	Flash 4 bits	b5		b4		b3	b2	b1	0	3b Offset	
	RA	Flash 4 bits	b5		b4		b3	b2	b1	0	0	

(a)

(b)

Fig. 3. (a) STG1 DAC partitioning across STG1 phases. (b) Timing details of the implemented two-step SAR ADC.

Section IV presents the measured results of the prototype ADC, and Section V concludes this article.

II. System-Level Details

The two-step SAR ADC (Fig. 2) consists of a 9-bit first stage (STG1) and 13-bit second stage (STG2) with an interstage gain of 64× from the residue amplifier (RA). The first 4 MSBs of STG1 are resolved by a coarse flash ADC. Owing to redundancy, the ADC yields a total of 18-bit quantization. The input signal spans 6.6-Vppdiff and 3.3-V/0-V references are used for $V_{\text{refp}}/V_{\text{refn}}$.

STG1 DAC partitioning is shown in Fig. 3(a). The input is sampled onto 512 units of STG1 DAC followed by STG1 conversion. Once completed, the remaining residue is transferred and amplified onto STG2 by the RA followed by STG1 DAC reset. STG2 is a conventional SAR with two phases: sampling and conversion. Both STG1 and STG2 SAR logic use asynchronous timing for bit decisions and dynamic logic cells for a minimum digital logic delay similar to [12].

As shown in Fig. 3(b), the conversion time of STG1 consists of 21 ns for STG1 (Flash+SAR) conversion, 21 ns for residue amplification, and 5 ns for DAC reset, which adds up to a fixed conversion time of 47 ns independent of the input clock rate. The remaining time period is allocated to input sampling, which translates to 20 ns of sampling time at 15 MS/s.

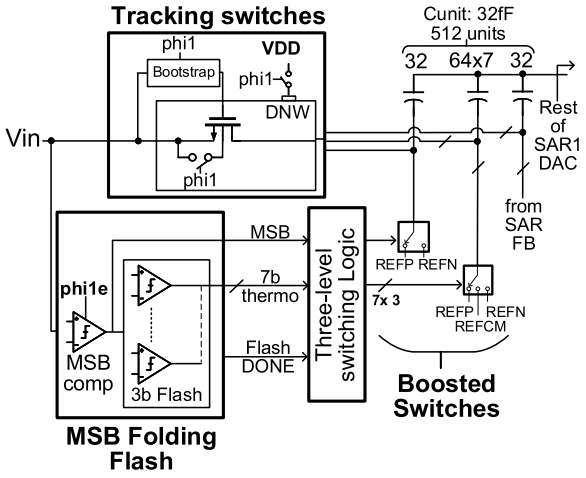
A. Flash/Three-Level Switching Scheme

The 4-bit MSB decisions are applied simultaneously with the bottom-plate common-mode (CM) using a coarse 4-bit MSB-folding flash ADC [13], as shown in Fig. 4(a). Three main advantages of using a coarse ADC are: 1) reduced STG1 conversion time; 2) residue generated at SAR1 top plate is reduced allowing the use of low-voltage devices for top-plate switches and STG1 comparator inputs; and 3) reducing current drawn and reducing capacitive loading on the reference voltage. In addition, as shown in Fig. 4(a) and (b), the flash feedback uses three-level switching (V_{refp}, V_{CM}, V_{refn}), which further reduces the capacitive loading on the reference voltage when compared with a two-level switching scheme [14]. For two-level switching, the maximum capacitive load seen is at mid-code, where half of the DAC capacitance is connected to V_{refp} in series with the other half to V_{refn}. For three-level switching, by using V_{CM}, reference is loaded by only $1/16$ of the DAC at mid-code. The MSB-folding architecture reduces the input-signal loading and power consumption, since it uses only eight comparators to resolve 4 bits compared with a conventional 4-bit flash using 15 comparators. The 4-bit MSBs of STG1 DAC are partitioned into one 32-unit and seven 64-unit capacitors, which conveniently match the MSB-folding flash ADC output (MSB + 7-thermometer LSBs) with the three-level switching scheme with a minimal logic overhead.

B. Input Sampling

Fig. 4(a) shows the details of the input-sampling switches. To maintain high linearity for the high input-signal frequencies, the bottom-plate tracking switches are bootstrapped [15]. Parasitic nonlinear junction capacitances are reduced by having the bulk track the input. In addition, the tracking switches are kept in a separate deep N-well (DNW), which is floating during the tracking phase as in [16]. In order to minimize the junction capacitances of the reference switches that share the same bottom-plate nodes, the switches use boosted gate controls that allow them to be minimally sized nMOS switches.

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(a)

Flash Section Three-Level Switching									
4-bit Flash Code	MSB	3-bit LSB (7 units thermometer)							
	32	64	64	64	64	64	64	64	
15	1	1	1	1	1	1	1	1	
14	1	1	1	1	1	1	1	0	
…	…	…	…	…	…	…	…	…	…
9	1	1	0	0	0	0	0	0	
8	1	0	0	0	0	0	0	0	
7	-1	0	0	0	0	0	0	0	
6	-1	-1	0	0	0	0	0	0	
…	…	…	…	…	…	…	…	…	…
1	-1	-1	-1	-1	-1	-1	-1	0	
0	-1	-1	-1	-1	-1	-1	-1	-1	

1	VREFP
0	VCM
-1	VREFN

(b)

Fig. 4. (a) Simplified block diagram of MSB-folding flash in addition to the details of input signal-tracking switches. (b) Mapping of three-level switching from flash output to STG1 DAC.

Finally, the top-plate sampling switches are also boosted and use 1.8-V devices for wide tracking bandwidth.

C. Dither

Dither is added to randomize the residue generated by STG1 in order to decorrelate the residue from the input signal. This spreads harmonic content due to nonlinearity of subsequent stages (RA, SAR2) into the noise floor and improves the spurious-free dynamic range (SFDR) performance of the ADC. A 4-bit dither code is generated from a 2^{15}-sequence length linear-feedback shift register (LFSR) and sampled onto a portion of STG1 DAC during the sampling phase. The redundancy allocated between the flash section and the SAR section of STG1 DAC accommodates the dither signal in addition to any flash ADC errors. The dither signal is subtracted digitally at the output of the ADC. While the dither provides, to some degree, a shuffling effect for the SAR section of STG1, it is more important for providing dithering to the RA and the backend stage, STG2. For the dither to propagate into the backend stages, special odd scaling of the dither reference voltage is implemented as in [11]. The residue curves, shown in Fig. 5, illustrate, for example, the output residue of SAR1 generated for a 4-bit dithered input. For a dither reference voltage scaled with an even fraction (1/2), the 15-input dither levels generate only four output residue levels by the end of SAR1 conversion. On the other hand, by scaling the reference with an odd fraction ($1/2 \times 17/16$), 15 unique levels of residue remain by the end of SAR1 conversion.

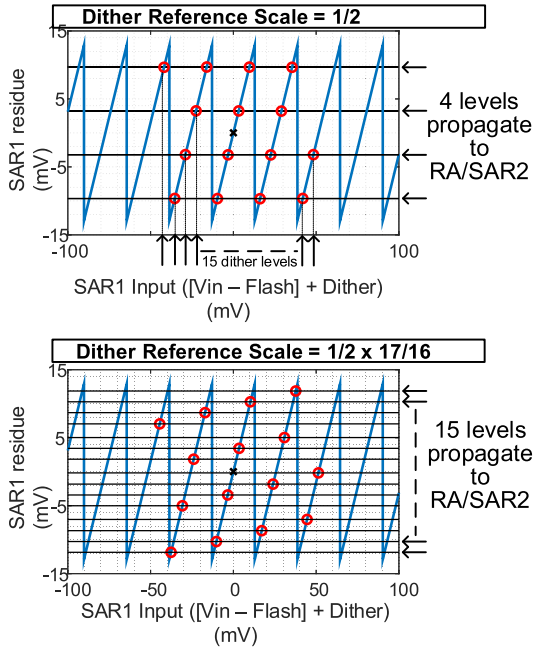


Fig. 5. Effect of dither reference scaling on propagation of dither to the RA and backend STG2 [11].

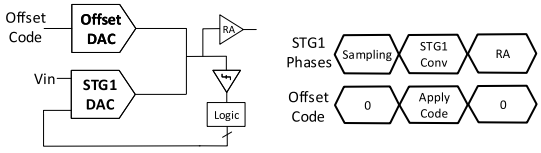


Fig. 6. SAR1 comparator offset trimming: implementation and timing.

D. Offset Trimming

SAR1 comparator offset shifts the residue at the end of SAR1 conversion into the redundancy range of the backend SAR2 ADC, which can easily be comparable in magnitude to the residue itself. To mitigate this effect, a one-time offset trim is performed by a 3-bit offset trim DAC sharing the SAR1 DAC top plate, as shown in Fig. 6. The DAC is set to V_{CM} during sampling, and then, the offset code is applied during SAR1 conversion. At the end of SAR1 conversion, the residue at the top plate consists of $\pm 1/2$ LSB residue in addition to the SAR1 comparator offset. The offset DAC is then reset to V_{CM} prior to residue amplification, which sub-

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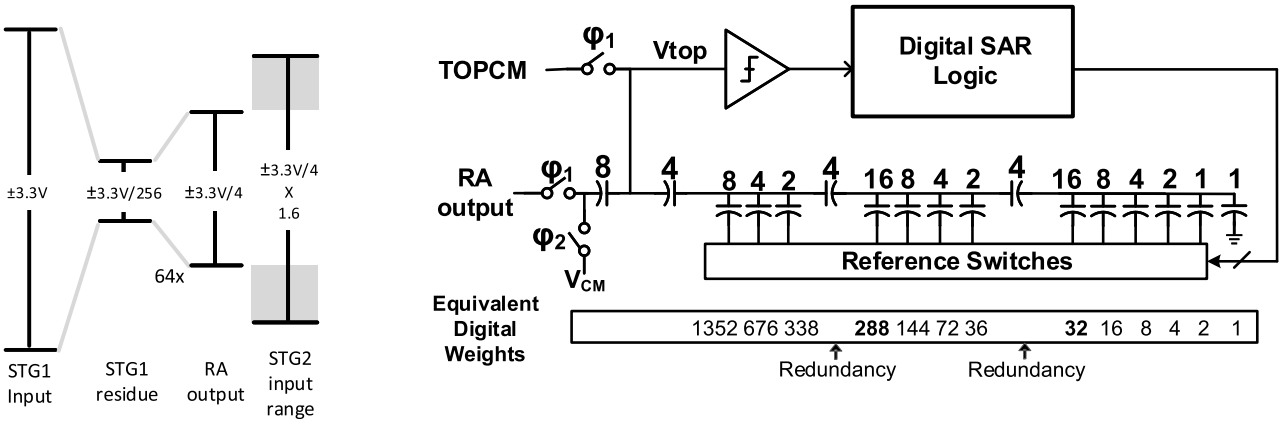


Fig. 7. STG2 DAC implementation with the illustration of signal-level progression from the input signal to the STG2 input.

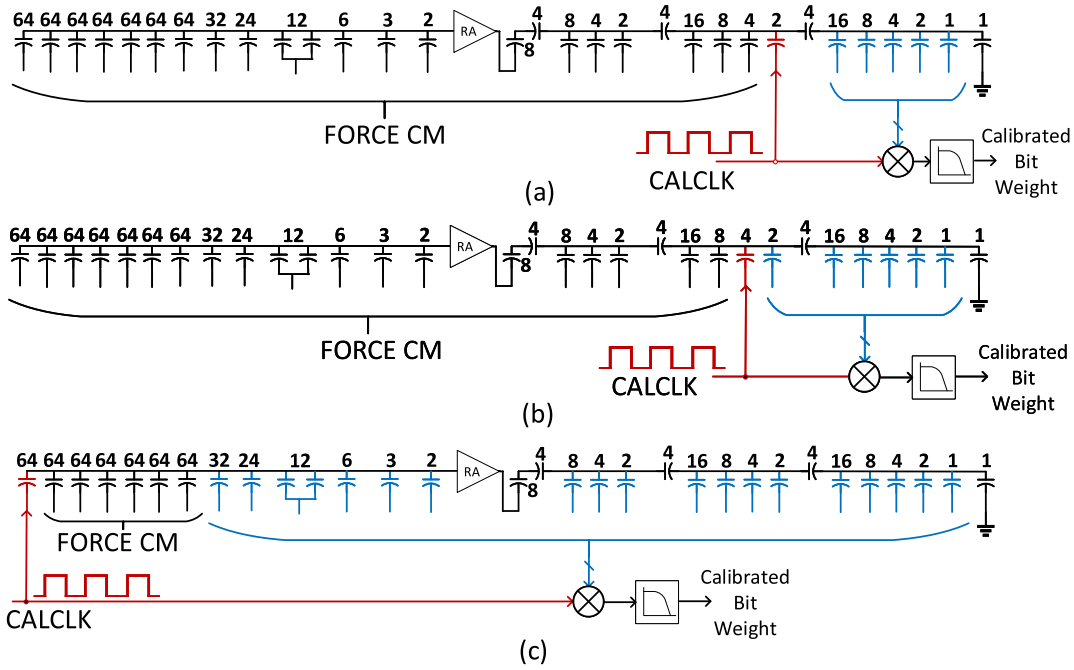


Fig. 8. Bit-weight detection method for (a) seventh LSB of STG2, (b) eighth LSB of STG2, and (c) MSB of STG1. CALCLK toggles the DAC weight under calibration between Vrefp and Vrefn (Red). Capacitor units in blue are used in SAR operation to measure the weight of the DAC unit under calibration.

tracts out the comparator offset. By using three-level switching and single-ended switching, two extra bits of resolution is added to the offset DAC resulting in a step size of 1/4 LSB.

E. Stage-2 SAR

Fig. 7 shows the architecture of STG2 ADC with three bridged sub-DACs. Redundancy is introduced between each sub-DAC allowing for shorter bit cycle durations for MSB decisions. Since STG1 resolves the input to 8-bit level ($FS/256$), an RA gain of 64× amplifies the residue back at the input of SAR2 to $FS/4$. Smaller reference voltages can be used to optimize the input range of SAR2 to the $FS/4$ RA output. Instead, the same STG1 references are used and the scaling down of the reference is obtained through the ratio of the 8-unit input sampling capacitor to the effective series combination of the first 4-unit bridge capacitor and the STG2 DAC, leaving the input range of STG2 60% larger than the amplified STG1 residue.

Since STG2 performs 13 bit decisions, STG2 design focuses on minimizing the bit-cycle duration. In addition to using dynamic logic cells, two techniques were employed to further speed up the conversion time: dynamic digital supply and impedance-matching, which were described in [17].

F. Bit-Weight Calibration

On-chip bit-weight detection allows for configuring each DAC capacitor in both STG1 and STG2 for foreground calibration [18]. The DAC capacitor weights are then used for off-chip bit-weight digital correction. Calibration is performed by toggling the DAC bit under calibration between V_{refp} and V_{refn}, while forcing the more significant bits to CM voltage, as shown in Fig. 8. The weight of the toggled DAC bit is

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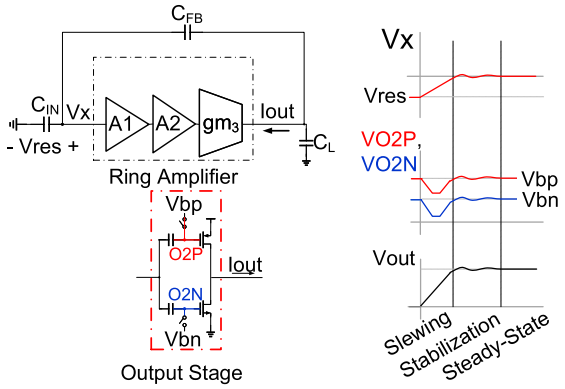


Fig. 9. RAMP basic structure and typical transients at various nodes during the residue amplification phase.

then measured with the following LSBs by demodulating and averaging the DAC weight. This process starts with the seventh LSB of STG2 [Fig. 8(a)] and proceeds until the weight of the MSB of STG1 is calibrated. The 6 LSB bits of STG2 are assumed to be sufficiently linear. Care must be taken to prevent offsets and thermal noise from saturating the dynamic range of the LSBs measuring the toggled DAC weight. This is achieved by the introduced redundancy between the sixth and the seventh LSB bits of STG2. The on-chip bit-weight detection logic allows for the DAC reference switches to be overridden performing the mentioned calibration procedure.

III. Residue Amplifier

The RA plays a critical role in the overall performance of the two-step SAR ADC. The residue amplification should meet the 18-bit quantization target in terms of linearity. The input-referred thermal noise of the ADC is dominated mainly by the input sampling kT/C noise and the thermal noise sampled onto SAR2 input during residue amplification phase, where the noise performance of the RA comes in.

A. Ring-Amplifier Overview

Fig. 9 shows the basic structure of the RAMP configured in the residue amplification phase with C_{IN} and C_L representing STG1 DAC and STG2 input-sampling capacitor, respectively [3]. The RAMP consists of a three-stage amplifier with a dominant pole at the output: two wide-bandwidth stages (A_1, A_2) followed by the output stage.

The RAMP large-signal transients are related to the large gain of A_1 A_2 preceding the push–pull output stage (Fig. 9). The deadzone bias usually refers to the voltages, Vbp and Vbn, separately biasing the pMOS and nMOS devices of the output stage, which sets the quiescent current and g_{m3} at steady state.

1) RAMP Transients: At the start of the RA phase, the residue is present at the input of the RAMP, V_x, and is quickly amplified by the first two stages to the inputs of the output stage, $O2P$ and $O2N$, turning on strongly the appropriate output device. The large output current charges the load capacitance quickly in the phase known as "slewing." As the output approaches the steady-state value, the RAMP transitions from the large-signal behavior into steady state during the "stabilization" phase. At "steady state," the residual transients settle, and the conventional ac analysis determines the final settling accuracy and the noise performance of the amplification phase.

2) RAMP Noise Considerations: For low-noise RAMP design, special attention to the small-signal parameters of the RAMP is required. For that purpose, the case of the single-stage operational transconductance amplifier (OTA) configured as the RA is first examined and later compared with that of the RAMP. The output-noise power for a single-stage OTA, $v^2_{\text{out,OTA}}$, during residue amplification was derived in [19] and shown in Fig. 10(a), where $\beta = C_{fb}/(C_{fb} + C_{IN})$, $C_{\text{eq}} = C_L + (1-\beta)C_{fb}$, and α represents the excess noise factor that captures noise added by OTA devices such as active loads and so on. While the input-noise power spectral density (PSD), $\overline{v_i^2}/\Delta f$, is inversely proportional to the transconductance of the OTA, g_{m1}, the output noise power, $v^2_{\text{out,OTA}}$, is independent of g_{m1}. This is because the effective noise bandwidth (NBW) is proportional to g_{m1} and cancels out with the $1/g_{m1}$ term in the noise PSD during integration.

An additional ratio, $g_{m,\text{eff}}/g_{m1}$, appears in the output-noise power for the case of the RAMP [Fig. 10(b)], where $gm_{\text{eff}} = A_1\ A_2\ g_{m3}$. This is due to the fact that the NBW of the RAMP is proportional to gm_{eff}. Therefore, while gm_{eff} is chosen to achieve a target steady-state bandwidth, the ratio, $g_{m,\text{eff}}/g_{m1}$, must be considered for noise. It is interesting to note that if the ratio is designed to be equal to 1, i.e., g_{m1} is chosen to be equal to gm_{eff}, then both the single-stage OTA and the RAMP achieve the same steady-state noise and small-signal gain bandwidth (GBW_{ss}). Yet, the RAMP can afford to have a lower GBW_{ss} than a single-stage OTA due to the faster settling behavior exhibited by the nonlinear dynamics of the RAMP. In addition, the small swing at the output of the first stage of RAMP allows for using inverter-based structures for current reuse, maximizing gm_1 for a given current. The excess noise factor, α, for an RAMP is approximately 1, since the first stage is free from any active loads and the noise sources of the second and third stages are divided by the gain of A_1 and $A_1\ A_2$, respectively. By designing for a low gm_{eff} that satisfies the target settling time and a large gm_1 for noise, good noise performance with low power consumption can be achieved.

B. Dual-Deadzone Ring Amplifier

The steady-state settling and noise performance of the RAMP depend on a well-defined steady-state set by g_{m3}.[1] For the typical output stage shown in Fig. 9 referred to as "Single-Deadzone" output stage [5], two design parameters' set g_{m3}: sizing of the output-stage devices and the chosen deadzone bias. Fig. 11(a) shows the I–V characteristics and large signal Gm-V characteristics for a small-sized and a large-sized output stage, both biased appropriately for the same steady-state g_{m3}, i.e., at V_x equals to 0. The input range across which g_{m3} is

[1]The location of the nondominant poles of the first stage and the second stage also plays an important role in the settling dynamics, as discussed later in this section.

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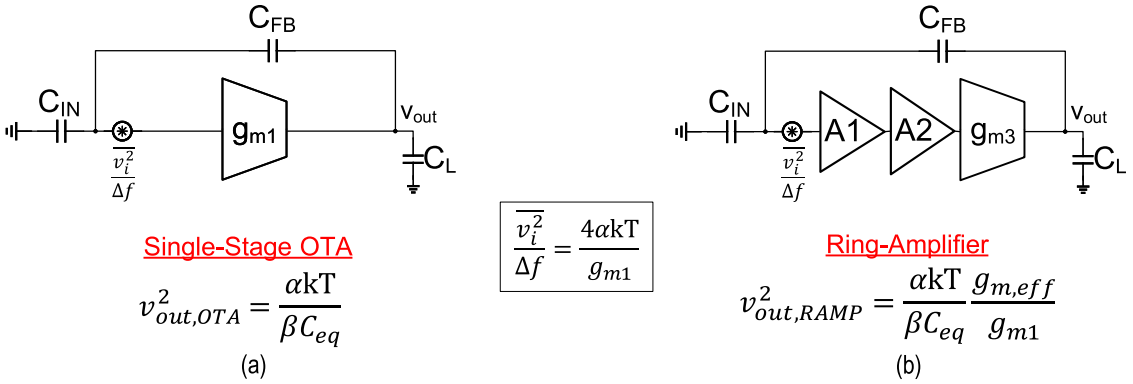


Fig. 10. Circuit model for noise analysis for the case of (a) single-stage OTA [19] and (b) RAMP.

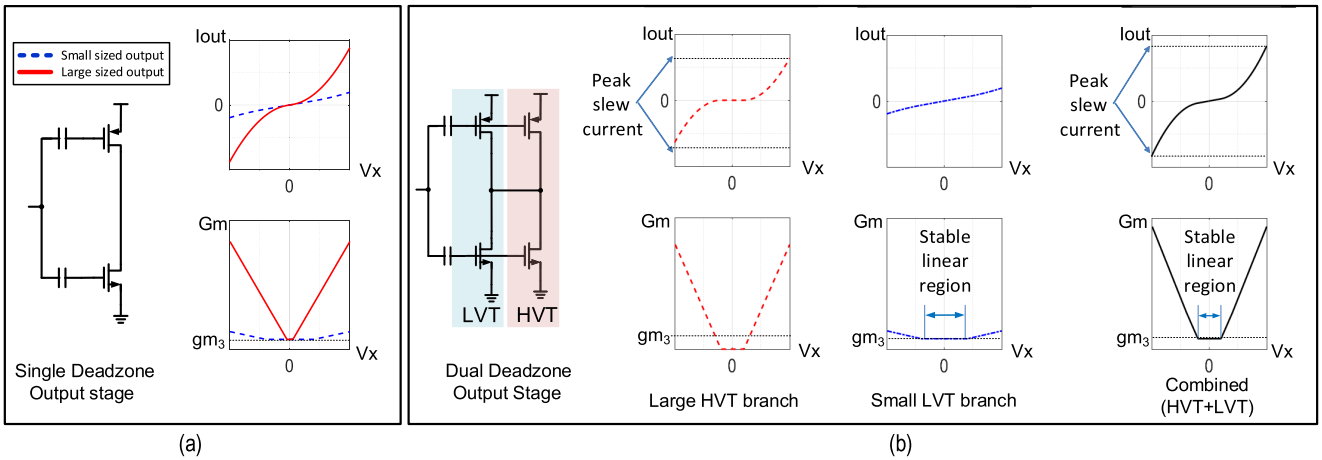


Fig. 11. I–V characteristics of the output stage for (a) small/large-sized single-deadzone output stage and (b) DDZ output stage.

constant is referred to as the "stable linear region" (SLR), where amplifier transients are dictated by the steady-state region. For the same target g_{m3}, a small-sized output stage would be biased at a larger overdrive voltage than a large-sized output stage. The larger overdrive voltage implies that the output stage maintains the steady-state g_{m3} over a wider input range [Fig. 11(a)], i.e., a wider SLR. A wide SLR implies a shorter stabilization phase. In addition, a large overdrive voltage implies lower sensitivity to the deadzone bias.

The disadvantage of the small-sized output stage is a slow-slewing phase due to the low peak current provided. In other words, the small-sized output stage acts for the greater part as a linear amplifier. On the other hand, the large-sized output stage provides a large peak slew current at the expense of a very narrow SLR, which prolongs the stabilization phase. On top of that, the large-sized devices would be biased in deep subthreshold to achieve the target gm_3, which results in high sensitivity to the deadzone bias.

In this work, the DDZ output stage shown in Fig. 11(b) is proposed as a solution that provides an additional degree of freedom, breaking free from the tradeoffs discussed above. The DDZ output stage consists of two branches, a small-sized low-threshold devices (LVT) branch and a large-sized high-threshold devices (HVT) branch. The HVT branch provides large slew current. The deadzone bias is such that near the steady state, the HVT branch shuts off and the small-sized LVT branch defines g_{m3} and, hence, the SLR. The width of the SLR and the peak slew current can be designed separately with the three design parameters: LVT branch sizing, HVT branch sizing, and the deadzone bias. The concept of having two separate paths in parallel, one that provides the slewing current and another for the fine settling of the output, was first introduced in RAMP-based ADCs in [5]. The implementation in [5] uses two separate RAMPs in parallel: a coarse RAMP for slewing and a fine RAMP for settling. In this work, the DDZ output

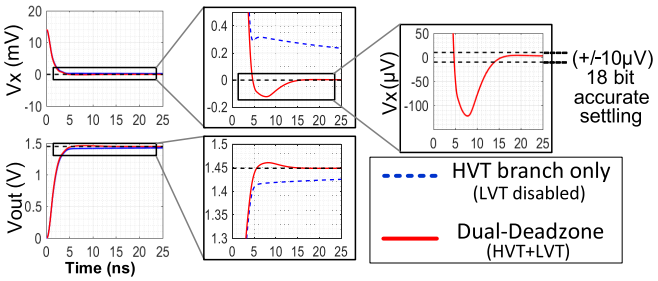


Fig. 12. Transient simulation illustrating settling characteristics of the DDZ output stage.

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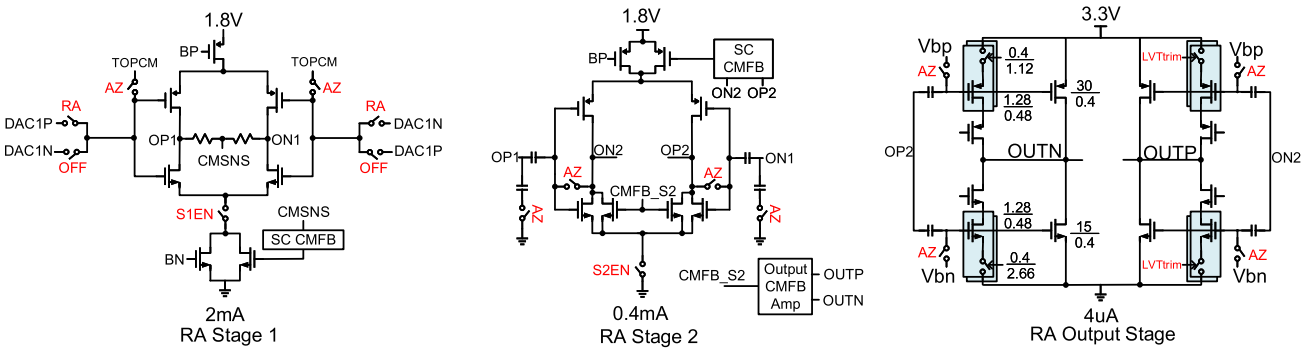


Fig. 13. Schematic details of the RA.

stage provides the same functionality with a minimal circuit overhead but at the cost of additional loading on the second stage of the RAMP. The location of the nondominant poles of the first stage and second stage of the RA also plays an important role in the settling dynamics. Therefore, in addition to the explanation provided using the static I–V characteristics of the output stage (Fig. 11), the extra loading imposed by the DDZ output stage on the second stage of the RAMP must be considered as to not degrade the stability of the RAMP at steady state.

Transient simulation waveforms at the input (V_x) and output (V_{out}) of the RAMP for a full-scale residue step are shown in Fig. 12 for two cases: 1) HVT-only and 2) DDZ. For both the cases, the output ramps up strongly toward the dotted steady-state value. As V_x approaches 300 μV away from 0, the HVT branch shuts off, suddenly slowing down for the HVT-only case. For the DDZ case, the LVT branch takes over the residual transients and settles the input to less than ± 10 μV (18-bit accuracy).

C. Ring-Amplifier Circuit Details

A detailed schematic of the proposed RAMP is shown in Fig. 13. The first stage is comprised of a complementary differential pair, which consumes 2 mA of bias current from a 1.8-V supply. Common-mode feedback (CMFB) is implemented locally for the first stage. An extra pair of dummy off switches at the input provides cancellation to the differential activity at SAR1 DAC top plate, since SAR1 conversion and RAMP autozeroing (AZ) occur simultaneously. AZ noise-filtering capacitors are added to the output of first stage as in [8]. The second stage, which consumes 0.4 mA from 1.8-V supply, similarly consists of a complementary differential pair with one tail current source for supply rejection and proper definition of current drawn. Global CMFB ensures that the amplifier output CM is set to the desired CM level. The output CM level is fed back into the second stage through a CMFB amplifier. This CMFB loop comprises the second stage, output stage, and CMFB amplifier. The second stage presents a nondominant pole to this loop. In order to improve the stability of the global CMFB loop, an additional local switched-capacitor (SC) CMFB loop is added to the second stage, which provides a reduced CM resistance at the output of the second stage, extending the nondominant pole there in addition to reducing the gain and gain-bandwidth of the global CMFB loop. Finally, the DDZ output stage consumes 4 μA from a 3.3-V supply. The higher supply allows for abundant swing with negligible static power consumption penalty. In addition, the LVT branch is digitally programmable (LVTtrim) from 1 to 3$\times$ for programmability of g_{m3}. The LVTtrim switches use long channel-length devices to provide degeneration to the LVT branch which desensitizes the gm_3 from the deadzone bias. This allows the deadzone bias to tune the width of the SLR for roughly the same g_{m3} value. The LVT branch is cascoded for higher steady-state dc gain without affecting the slewing of the HVT branch. The deadzone bias, V_{bn}/V_{bp}, is generated using LVT current mirrors to set the quiescent current in the LVT branches. The amplifier obtains 25/25/60 dB from the first/second/third stage for a 110-dB total dc gain across the output swing of the RA, which satisfies the 18-bit linearity requirement.

Power duty-cycling of the RAMP is essential for power saving. Prior to the start of amplification (during SAR1 conversion), the RAMP turns on and uses autozeroing to set the bias voltages of the internal nodes in addition to rejecting low-frequency flicker noise and offset. Fig. 14 shows a simplified version of the RAMP to illustrate the power up and autozeroing sequence prior to amplification. For power savings, it is desirable to reduce both the AZ and the RA time durations, i.e., a fast power-on plus AZ sequence followed by a fast amplification phase. Using short AZ durations implies incomplete settling across the AZ capacitors. In order to achieve proper AZ operation, the main strategy is to ensure that the previously stored voltages across AZ capacitors are preserved from sample to sample. Hence, AZ settles across multiple cycles. Special precautions are taken to ensure that AZ capacitors do not leak during the power-down period and during the power-on sequence.

First, BP/BN provides the tail current source biases for the first two stages and the CMFB amplifier. During power-on, a large voltage step occurs at the drain of the tail current sources, $V_{\text{dp}}/V_{\text{dn}}$, that feeds through the C_{gd} of those devices back onto BP/BN. For fast settling of BP/BN, a large bias current mirror would be needed wasting unnecessary power. Instead, BIASEN switches shown in Fig. 14 disconnect the BP/BN node from the bias circuit prior to power off. As shown

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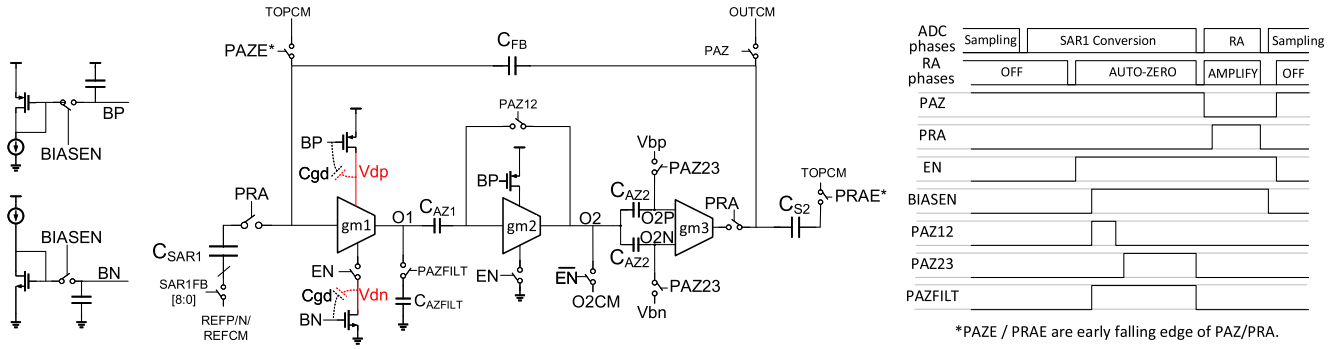


Fig. 14. RAMP simplified single-ended representation. Schematic and timing details are shown for power duty-cycling and auto-zero operation.

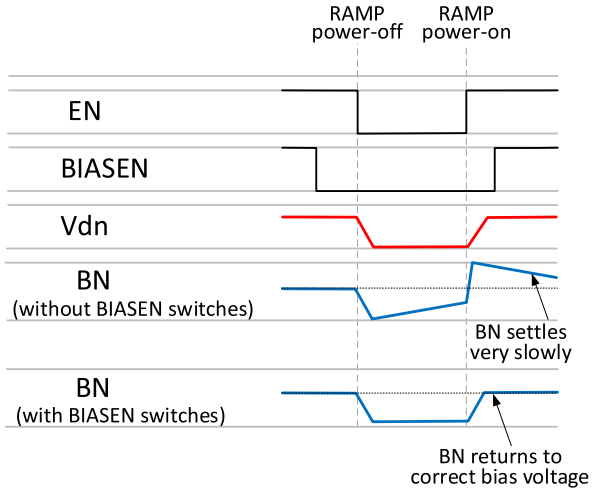


Fig. 15. Waveforms of the voltage nodes, V_{dn} and BN, during power duty-cycling illustrating the importance of BIASEN switches of Fig. 14 for providing fast bias recovery. V_{dp} and BP behave in a similar fashion.

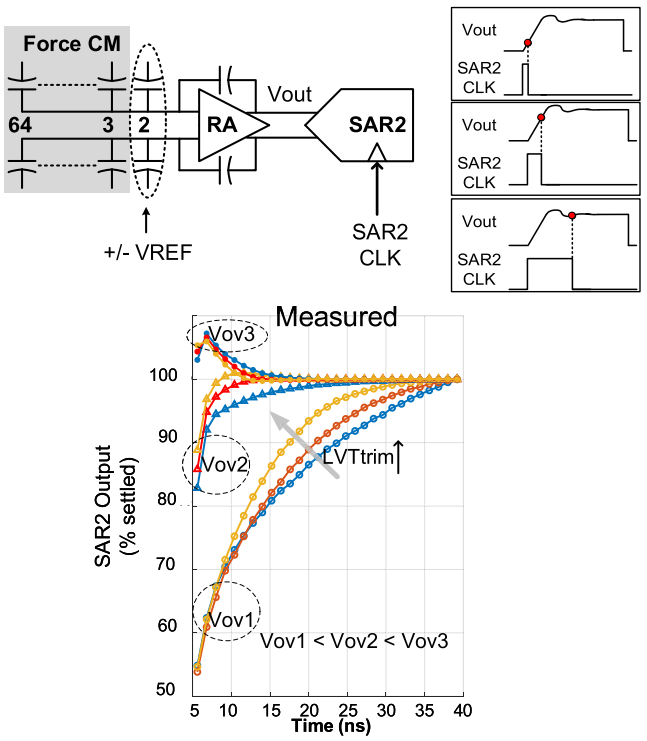


Fig. 16. Method for on-chip RA settling characterization.

in Fig. 15, as the circuit powers off and then powers on again for the next cycle, the drain voltages, V_{dp}/V_{dn}, start and end at the same voltage causing a net zero charge feedthrough at BP/BN, after which the BIASEN switches are turned on again. Second, the AZ phase is delayed from the power-on (EN) signal to allow the internal nodes to recover to their previous levels prior to any AZ updates. Third, AZ between stages 1/2 (PAZ12) and between stages 2/3 (PAZ23) are nonoverlapping, which allows the second stage to charge C_{AZ1} during PAZ12 without being loaded by C_{AZ2}.

Finally, care must be taken to avoid forward-biasing any switch junction that would leak the stored AZ voltages during power off. The CM voltage at O2 during normal operation is VDD/2, while O2P/N are a mere V_{gs} away from their respective supply sides. During power off, O2 is floating and could drift to VDD or VSS, forward-biasing the drain-bulk junctions of the PAZ23 switches. Hence, to eliminate this issue, a replica bias voltage (O2CM) is connected to O2 during power off to hold the voltage at O2 in place.

D. Ring-Amplifier On-Chip Characterization

An on-chip method was devised for observing the RAMP settling response (Fig. 16). The LSB unit of SAR1 is toggled to $\pm V_{ref}$ every other cycle while the remaining elements are forced to CM. SAR2 captures the RA output at the SAR2 CLK falling edge. By having a programmable SAR2 CLK width from 5 to 40 ns with 1-ns step, the RA output is sampled and averaged at each of those sampling instances to reconstruct the settling response of the RAMP. The measured settling response can be analyzed across various settings for insight into the settling behavior of the RAMP. V_{ov} represents different deadzone bias settings, and for each deadzone bias setting, the settling behavior is captured for the three LVTtrim settings.

A sample of such an insight is shown in Fig. 17, where the settling time to 18-bit accuracy is plotted against the deadzone bias for the three LVTtrim settings. While the units for the deadzone bias are arbitrary, a larger number indicates a larger V_{gs}, and therefore, a smaller SLR. For a large SLR (small x-axis value), the amplifier operates mainly in steady state as it transitions from slewing to steady state too early.

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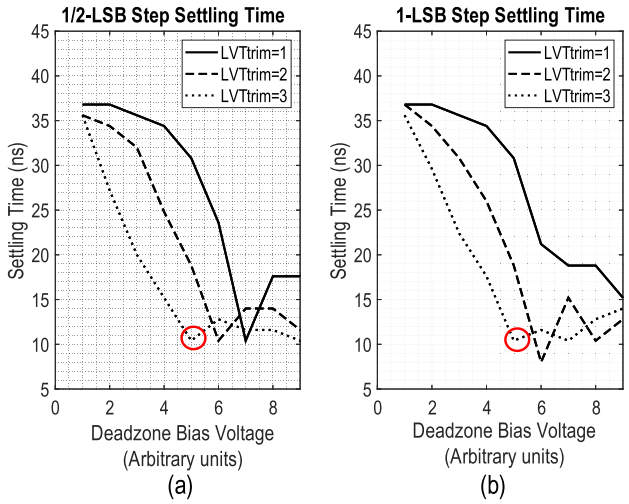


Fig. 17. Settling time across various deadzone bias voltages and LVTtrim settings for (a) 1/2-LSB and (b) 1-LSB input residue step. 1-LSB refers to two cap units of SAR1 DAC. Both x-axes are scaled to the same reference.

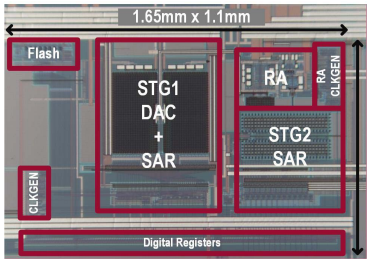


Fig. 18. Die photograph.

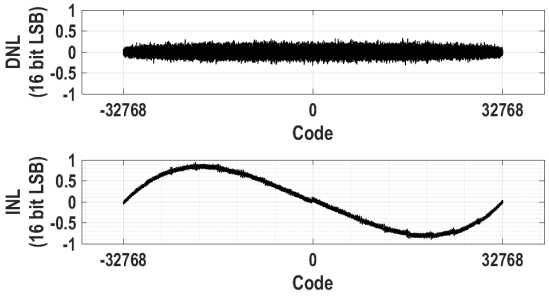


Fig. 19. Differential/Integral nonlinearity (DNL/INL) characteristics.

As SLR decreases, settling time improves. The large signal delay through the first two stages plays a role too in the settling time of the RAMP. Even when the virtual ground (V_x) has entered the SLR, the shut-off of the HVT branch is delayed by the first two stages. At a certain deadzone bias, a settling time minima occurs when such delay pushes the output extremely close to the expected steady-state value for the given residue input. Hence, it is important to observe the settling time for different residue input levels, as shown in Fig. 17. The circled deadzone setting was used for the data collection of the measurement results in Section IV. Other parameters that are not shown here such as steady-state gain error and noise performance can also be analyzed. In addition, multiple residue steps can be generated by using the input dither with SAR1 operation, as shown in Fig. 4.

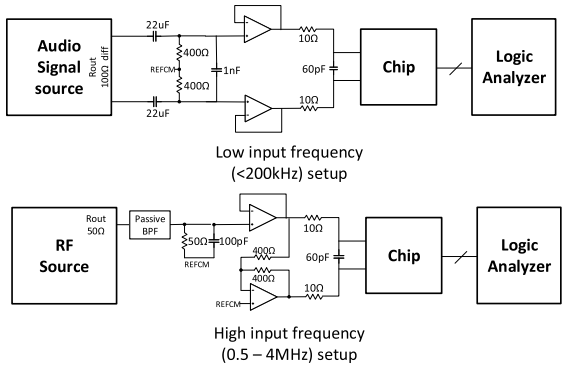


Fig. 20. Dynamic performance measurement setup for low- and high-frequency inputs.

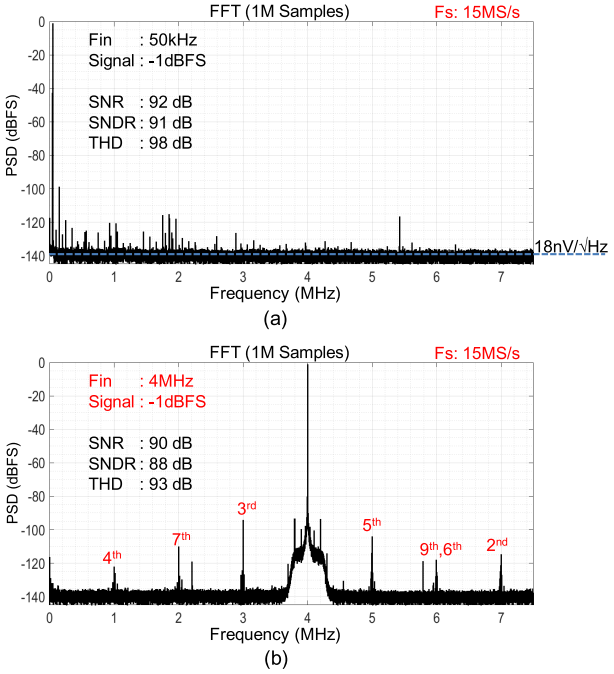


Fig. 21. PSD at −1 dBFS for (a) 50-kHz input signal and (b) 4-MHz input signal, both at 15 MS/s.

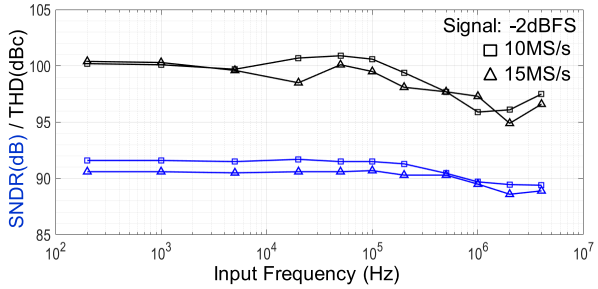


Fig. 22. SNDR/total harmonic distortion (THD) versus input signal frequency sweep at −2 dBFS for 10 and 15 MS/s.

IV. Measurement Results

The proposed ADC is fabricated in 0.18-μm 1.8-V/3.3-V CMOS technology and occupies an area of 1.82 mm^2 (Fig. 18). The 16-bit ADC achieves an integral non-linearity (INL) of ±1 LSB, as shown in Fig. 19. Fig. 20 shows

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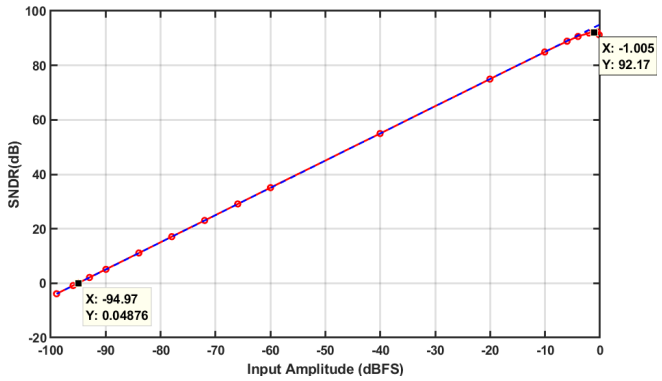


Fig. 23. Dynamic range measurement at 2 MS/s with 1-kHz input showing a dynamic range of 95 dB.

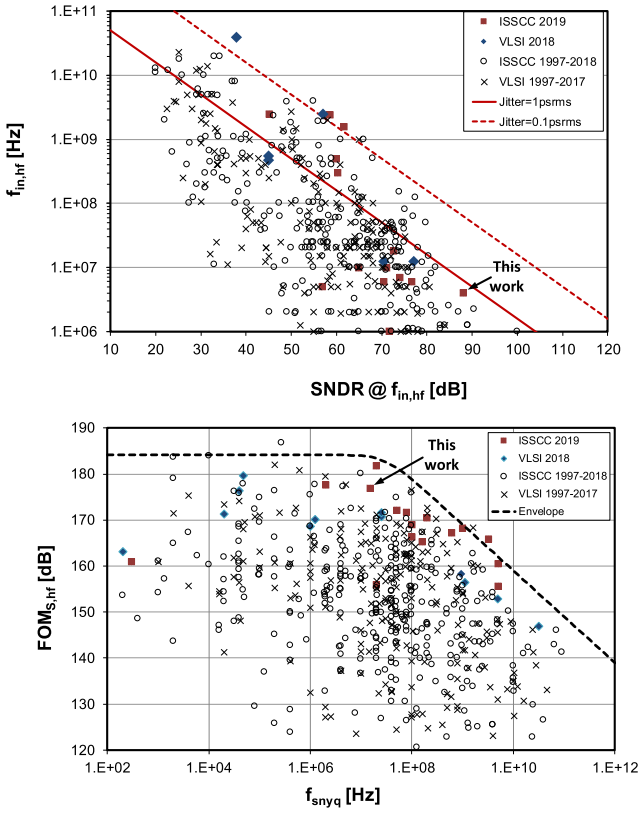


Fig. 24. Input frequency versus SNDR and Schreier figure-of-merit (FOM) charts from data in [20].

the measurement setup for low- and high-frequency inputs. An audio source with a differential output was used for low-frequency testing up to 200 kHz with two single-ended unity-gain buffers driving the ADC. The filter networks before and after the buffers are tuned carefully for an optimized "noise"-"settling time" tradeoff at each tested sampling rate. The ADC achieves a peak SNDR of 91 dB at a −1 dBFS 50-kHz input [Fig. 21(a)]. For input frequencies higher than 200 kHz, a single-ended RF source followed by a passive bandpass filter is required to achieve 16-bit linearity. The output is then transformed to a differential signal using a combination of a unity-gain buffer and an inverting-gain buffer. The ADC was tested up to 4 MHz (due to unavailability of bandpass filters with higher center frequencies) and achieves 88-dB SNDR at −1 dBFS. The increased noise floor and tones around the fundamental [Fig. 21(b)] are inherent to the RF source, and do not appear outside the bandpass filter bandwidth. Fig. 22 summarizes the SNDR/THD of the ADC across input frequencies for −2-dBFS input at 10 and 15 MS/s. Improvements at 10 MS/s are due to optimized noise filtering and tracking time for the input network. For dynamic range measurement, the ADC is tested by directly using the audio source without ADC drivers for lower input noise. At a sample rate of 2 MS/s with a 1-kHz input signal, the ADC achieves a dynamic range of 95 dB, as shown in Fig. 23.

The ADC consumes 9.8 mW at 15 MS/s, where STG1, RA, and STG2 consume 2.9, 2.7, and 2.2 mW, respectively. The clock generation and the references consume an additional 1 mW each. The ADC power scales linearly with the sampling rate due to the digital nature of the SAR blocks and, particularly, due to the power duty-cycling of the RA. A comparison with the state of the art is shown in Fig. 24.

V. Conclusion

This article presents the first RAMP-based ADC achieving an SNDR of 91 dB, where a DDZ RAMP is proposed to achieve the required settling accuracy and speed while benefiting from the inherent power efficiency of the RAMP structure. Furthermore, a method was devised for characterizing the settling response of the RAMP on-chip. The ADC achieves a high $\text{FOM}_{S,lf}$ of 179.8 dB and an $\text{FOM}_{S,hf}$ of 176.8 dB when compared with the state of the art.

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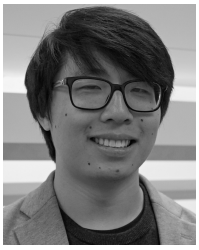
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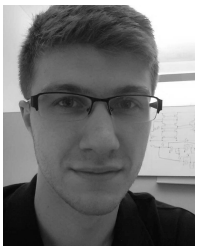
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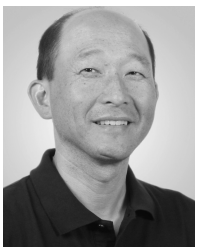
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